

Huaizhi Qu

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Research Interest

Machine Learning:

- Foundation models (e.g., Multimodal LLM, LLM)
- Efficiency for foundation models (e.g., LLM inference acceleration, model merging)
- Interpretability and explainable AI (e.g., uncertainty quantification)

Computer Vision: 3D Vision, Neural Rendering

Education

The Hong Kong University of Science and Technology

Sep. 2026 -

- PhD in AMC, [World Mind Lab](#) • [心象实验室](#)
 - Advisor: [Prof. Fangneng Zhan](#) and [Prof. Hongbo Fu](#)

University of North Carolina at Chapel Hill

Aug. 2024 - Dec. 2025

- MS in Computer Science

University of Science and Technology of China

Aug. 2018 - July 2024

- BE in Computer Science

Internship

- **Tencent** (Project UP | 青云计划) May 2026 -
- **ByteDance** Sep. 2025 - Apr. 2026
- **Honda Research Institute USA** May 2025 - Aug. 2025

Publication (* denotes equal contribution)

PETS: A Principled Framework Towards Optimal Trajectory Allocation for Efficient Test-Time Self-Consistency

- (α - β) Zhangyi Liu*, **Huaizhi Qu***, Xiaowei Yin*, He Sun, Yanjun Han, Tianlong Chen, Zhun Deng
- The Forty-Third International Conference on Machine Learning (ICML 2026)

(Be Cautious!) Bio-Foundation Models Are Not Yet Robust to Biological Plausible Perturbations and ML Transformations

- Jinhao Duan, Ruichen Zhang, Gengwei Zhang, **Huaizhi Qu**, Jie Peng, Sijia Liu, Tianlong Chen
- The Forty-Third International Conference on Machine Learning (ICML 2026)

Can Large Language Models Assess and Reframe Psychological Attribution: A Benchmark and Analysis

- Qiang Zhou, Hanzhen Zhu, Jiyang Liu, Pan Wang, Yufeng Tian, Rui Tu, **Huaizhi Qu**, Zhuoran Wang, Xin Hu, Lei Li, Tianlong Chen, Jingtong Hu
- The Forty-Third International Conference on Machine Learning (ICML 2026)

LucidAtlas: Learning Uncertainty-Aware, Covariate-Disentangled, Individualized Atlas Representations

- Yining Jiao, Sreekalyani Bhamidi, Carlton Jude ZDANSKI, **Huaizhi Qu**, Julia S Kimbell, Andrew Prince, Cameron P Worden, Samuel Kirse, Christopher Rutter, Benjamin H Shields, Jisan Mahmud, Tianlong Chen, Marc Niethammer
- Transactions on Machine Learning Research (TMLR)

MonoVLM: Monocular 3D Visual Grounding with Vision Language Models

- **Huaizhi Qu**, Hossein Nourkhiz Mahjoub, Vaishnav Tadiparthi, Kwonjoon Lee, Tianlong Chen
- The IEEE/CVF Conference on Computer Vision and Pattern Recognition 2026 (CVPR 2026)

VLM-3R: Vision–Language Models Augmented with Instruction-Aligned 3D Reconstruction

- Zhiwen Fan, Jian Zhang, Renjie Li, Junge Zhang, Runjin Chen, Hezhen Hu, Kevin Wang, **Huaizhi Qu**, Dilin Wang, Zhicheng Yan, Hongyu Xu, Justin Theiss, Tianlong Chen, Jiachen Li, Zhengzhong Tu, Zhangyang Wang, Rakesh Ranjan
- The IEEE/CVF Conference on Computer Vision and Pattern Recognition 2026 (CVPR 2026)

BetaConform: Efficient MAP Estimation of LLM Ensemble Judgment Performance with Prior Transfer

- **Huaizhi Qu**, Inyoung Choi, Zhen Tan, Song Wang, Sukwon Yun, Qi Long, Faizan Siddiqui, Kwonjoon Lee, Tianlong Chen
- The 39th Annual Conference on Neural Information Processing Systems (NeurIPS 2025)

Multi-Agent Debate for LLM Judges with Adaptive Stability Detection

- Tianyu Hu, Zhen Tan, Song Wang, **Huaizhi Qu**, Tianlong Chen
- The 39th Annual Conference on Neural Information Processing Systems (NeurIPS 2025)

Bit-Flip Error Resilience in LLMs: A Comprehensive Analysis and Defense Framework

- Yuhang Chen, Zhen Tan, Ajay Jaiswal, **Huaizhi Qu**, Xinyu Zhao, Qi Lin, Yu Cheng, Andrew Kwong, Zhichao Cao, Tianlong Chen
- The 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP 2025)

KeyTAR: Practical Keystroke Timing Attacks and Input Reconstruction

- Mufan Qiu, Lihuan Chuang, Dohhyun Kim, **Huaizhi Qu**, Tianlong Chen, Andrew Kwong
- The 47th IEEE Symposium on Security and Privacy (IEEE S&P 2026)

UQ-Merge: Uncertainty Guided Multimodal Large Language Model Merging

- **Huaizhi Qu**, Xinyu Zhao, Jie Peng, Kwonjoon Lee, Behzad Dariush, Tianlong Chen
- The 63rd Annual Meeting of the Association for Computational Linguistics Findings (ACL 2025 Findings)

HoloZip: High Hologram Compression via Latent-of-Latent Coding

- **Huaizhi Qu**^{*}, Yujie Wang^{*}, Ruichen Zhang, Hengyu Lian, Mufan Qiu, Samarjit Chakraborty, Henry Fuchs, Tianlong Chen, Praneeth Chakravarthula
- 2025 IEEE International Conference on Computational Photography (ICCP 2025)

RTGSSLAM: An Edge GPU Plug-in for Real-Time 3D Gaussian Splatting-based SLAM Systems

- Leshu Li, Jiayin Qin, Jie Peng, Zishen Wan, **Huaizhi Qu**, Ye Han, Pingqing Zheng, Hongsen Zhang, Tianlong Chen, Yang (Katie) Zhao
- The 58th IEEE/ACM International Symposium on Microarchitecture (MICRO 2025)

Omni-Recon: Advocating a Pretrain-Bake-Finetune Paradigm towards Foundation Neural Radiance Fields

- Yonggan Fu, **Huaizhi Qu**, Zhifan Ye, Chaojian Li, Yingyan Celine Lin
- The 18th European Conference on Computer Vision (ECCV 2024, Oral)

GPU Occupancy Prediction of Deep Learning Models Using Graph Neural Network

- Hengquan Mei^{*}, **Huaizhi Qu**^{*}, Jingwei Sun[†], Yanjie Gao, Haoxiang Lin, Guangzhong Sun
- The 25th IEEE International Conference on Cluster Computing (Cluster 2023)

In Submission (* denotes equal contribution)

From Features to Geometry: Holistic Depth Awareness for Monocular 3D Object Detection

- Zhihao Zhang, Abhinav Kumar, **Huaizhi Qu**, Tianlong Chen, Xiaoming Liu
- Submitted to The 40th Annual Conference on Neural Information Processing Systems (NeurIPS 2026)

Token is Matter: Post-training Scaling Laws for Multimodal Large Reasoning Model with Arbitrary Modality Combinations

- Jie Peng, Zhuoxuan Zhang, Gengwei Zhang, **Huaizhi Qu**, Hossein Nourkhiz Mahjoub, Vaishnav Tadiparthi, Kwonjoon Lee, Ehsan Moradi Pari, Tianlong Chen

GEM: 3D Gaussian Splatting for Efficient and Accurate Cryo-EM Reconstruction

- **Huaizhi Qu**, Xiao Wang, Gengwei Zhang, Jie Peng, Tianlong Chen

EditCast3D: Single-Frame-Guided 3D Editing with Video Propagation and View Selection

- **Huaizhi Qu**, Ruichen Zhang, Shuqing Luo, Luchao Qi, Zhihao Zhang, Xiaoming Liu, Roni Sengupta, Tianlong Chen

CryoNeRF: generalizable automated cryo-EM reconstruction using neural radiance field

- **Huaizhi Qu***, Xiao Wang*, Yuanyuan Zhang, Sheng Wang, William Stafford Noble, Tianlong Chen

DOGe: Defensive Output Generation for LLM Protection Against Knowledge Distillation

- Pingzhi Li, Zhen Tan, **Huaizhi Qu**, Huan Liu, Tianlong Chen

Loop Your CPU and SSD in: Enabling Sustainable and Accessible LLM Inference using Mixed-Precision and Multi-Level Caching

- Jie Peng, Zhang Cao, **Huaizhi Qu**, Zhenyu Zhang, Chang Guo, Yanyong Zhang, Zhichao Cao, Tianlong Chen

Award

- HKUST Postgraduate Studentship
- NeurIPS 2025 Scholar Award

Research Experiences

Graduate Student

Aug. 2024 - Dec. 2025

Advisor: [Professor Tianlong Chen](#) (Assistant Professor, Department of Computer Science, UNC Chapel Hill)

Undergraduate Research, EIC Lab, Georgia Tech

Jan. 2023 - March 2024

Advisor: [Professor Yingyan \(Celine\) Lin](#) (Associate Professor, School of Computer Science, Georgia Tech)

Undergraduate Research, Algorithm and Data Application Lab, USTC

Aug. 2022 - May 2023

Advisor: [Professor Guangzhong Sun](#) (Professor, School of Computer Science and Technology, USTC)

Professional Services

Reviews: NeurIPS 2026, ECCV 2026, CVPR 2026, CPAL 2026, ICLR 2026, NeurIPS 2025, ICML 2025, AAAI 2025, ICLR 2025, NeurIPS 2024